Layer 7: SAM Chemistry

Three-Component Probe-Thiol/C6FC/MCH Monolayer: Deposition, Stability, and Ca^{2+} Artefact

Eton iGEM 2026, Dry Lab | Layer 7 of 17-Layer Biosensor Pipeline | `sam_loading.py`, `sam_stability.py`, `calcium_artefact.py`

Layer 7 characterises the self-assembled monolayer (SAM) that immobilises the peptide probe on the gold electrode surface. Its primary output is the probe surface density Γ_p, which passes to Layer 8 (steric accessibility) and Layer 9 (surface kinetics). Secondary outputs are bounds on SAM degradation over the 60-minute assay window and a quantitative account of the Ca^{2+} artefact and its ratiometric cancellation.

Every parameter value must be confirmed from a paper physically present in `papers/layer_07/`; values without a paper source are explicitly flagged throughout.

I. Mixed SAM Architecture

Molecular components

The sensor electrode carries a three-component thiol monolayer. The probe-thiol is a 10-residue peptide substrate conjugated at its N-terminus to a cysteine residue that anchors the molecule to gold, and at its C-terminal lysine to a methylene-blue (MB) redox label:

$$\underbrace{\text{Ac-Cys}}_{\text{gold anchor}}\text{-PEG}_6\text{-}\underbrace{\text{SGKIPRTLTA}}_{\text{substrate}}\text{-PEG}_2\text{-}\underbrace{\text{Lys(MB)}}_{\text{signal label}}$$

Gold attachment is through the thiol of the cysteine residue (Cys-SH → Au-S-Cys). The N-terminal acetyl cap (Ac) prevents non-specific gold binding via the free amine. The PEG_6 block provides a flexible hydrophilic spacer between the electrode surface and the substrate; the PEG_2 linker conjugates MB to the Lys ε-amine. MMP-9 (or HNE on the secondary channel) cleaves within SGKIPRTLTA, releasing the MB-bearing C-terminal fragment into solution and producing a signal-off decrease in SWV peak current proportional to enzymatic activity (Layer 0, Section V).

A co-immobilised reference reporter, 6-ferrocenylhexanethiol (C6FC, Fc–$(CH_2)_6$–SH), provides an internal electrochemical standard. C6FC and MCH (HS–$(CH_2)_6$–OH) both present a C_6 alkanethiol anchor to the gold surface and pack together with uniform chain geometry; the probe-thiol anchors via the shorter Cys residue and extends a bulky PEG_6-peptide chain into solution above them. Packing compatibility of all three components in a mixed SAM must be verified on the fabricated electrode. The ferrocene (Fc) headgroup gives a reversible one-electron redox signal. Adjémian et al. (2010) [12] report a formal potential of ∼150–155 mV vs. SCE (≈ +0.20 V vs. Ag/AgCl) for an alkyl-ferrocene label on an analogous Fc-peptide-PEG monolayer on gold (p. 10350 of [12]); this value is expected to be close to the C6FC formal potential but must be confirmed directly on the fabricated device. The Fc signal is well positive of the MB reduction peak; the two peaks are resolved in a single SWV scan without additional switching (Section I.3). The MB formal potential in the assay buffer is not confirmed from any paper in `papers/layer_07/` and must be determined by cyclic voltammetry on the fabricated electrode.

The diluent is 6-mercapto-1-hexanol (MCH, HS–$(CH_2)_6$–OH). All three components are co-deposited simultaneously from a single mixed solution (one-pot deposition); MCH occupies gold sites not taken up by probe-thiol or C6FC, presenting a chemically inert surface to the wound fluid and preventing non-specific adsorption of wound proteins to the bare gold.

Close-packed monolayer density

The maximum surface density of a fully close-packed alkanethiol monolayer on gold, measured by reductive desorption coulometry, is [6]:

$$\Gamma_{\max} = 9.3 \times 10^{-10}\ \text{mol cm}^{-2} = 930\ \text{pmol cm}^{-2} \tag{I.1}$$

Love et al. report an independent cross-check of approximately 750 pmol cm^{-2} for densely packed alkanethiol monolayers on Au(111) [1]. Equation (I.1) is an upper bound. The probe-thiol cannot approach $\Gamma_{\max}$ due to the bulky peptide headgroup; the target probe coverage Γ_p = 2–5 pmol cm^{-2} is 0.2–0.5% of the close-packed limit.

The following assumptions underlie this section:

1. Probe-thiol, C6FC, and MCH are co-deposited from a single mixed solution; their relative surface coverages are governed by competitive Langmuir adsorption.

2. $\Gamma_{\max} = 930$ pmol cm^{-2} is an upper bound on total SAM packing; the actual MCH density is lower because probe-thiol and C6FC occupy a fraction of the surface.

C6FC reference reporter and ratiometric signal

C6FC (Fc–$(CH_2)_6$–SH) is co-immobilised with the probe-thiol at a molar ratio that is to be optimised on the fabricated electrode. Its function is threefold.

1. **Internal area reference.** The Fc signal I_{Fc} is proportional to the number of C6FC molecules on the electrode and is insensitive to MMP-9 activity, because C6FC contains no peptide substrate. I_{Fc} therefore tracks electrode area and surface loading, providing a normalisation factor that cancels run-to-run variation in electrode geometry or deposition yield.
2. **Fouling correction.** Non-specific protein adsorption attenuates both the MB and Fc signals by approximately the same factor (fouling layer acts as a diffusion barrier). Taking the ratio $I_{\mathrm{MB}}/I_{\mathrm{Fc}}$ suppresses this common-mode attenuation.
3. **Ratiometric readout.** The device signal is the ratio:

$$R(t) = \frac{I_{\mathrm{MB}}(t)}{I_{\mathrm{Fc}}} \tag{I.2}$$

where I_{Fc} is measured simultaneously with $I_{\mathrm{MB}}(t)$ in the same SWV scan. The normalised fractional cleavage is:

$$\frac{\Delta R}{R_0} = \frac{R_0 - R(t)}{R_0} = f_{\mathrm{cleaved}}(t) \tag{I.3}$$

This equality holds *because* C6FC is the non-cleavable internal control: C6FC carries no peptide substrate, so MMP-9 and HNE cannot reduce I_{Fc}, making I_{Fc} constant over the assay. Without a non-cleavable internal reference, $\Delta R/R_0$ would reflect both cleavage and any drift in electrode response, and the equality $\Delta R/R_0 = f_{\mathrm{cleaved}}$ would not hold in general (see assumption below).

The electron-transfer kinetics of Fc-terminated alkanethiol chains on gold are characterised in Smalley et al. (2003) [8]: the tunnelling decay constant through the C_6 alkyl bridge is $\beta = 0.94 \pm 0.06$ Å^{-1}, confirming that ET from Fc to gold through the C_6 chain of C6FC is fast relative to the SWV timescale and produces a sharp, well-defined peak. The probe-thiol bridge from Au to MB runs through Cys and the PEG_6 spacer (not through a C_6 alkyl chain); the tunnelling decay constant through PEG chains is different from alkyl chains and is not confirmed from any paper in `papers/layer_07/` (see Section IV and Limitation 5).

The following assumption underlies this subsection. MMP-9 cleaves SGKIPRTLTA but has no known activity against the Fc–$(CH_2)_6$–SH molecule, which carries no peptide substrate; I_{Fc} is therefore expected to be stable over the 60-minute assay window. This has not been confirmed by a control experiment on the fabricated electrode and should be verified before ratiometric calibration is finalised. The target C6FC surface density relative to probe-thiol density is not confirmed from any paper in `papers/layer_07/` and must be optimised empirically.

II. Deposition Protocol and Target Surface Density

Deposition protocol

Wcisło et al. (2021) [11] describe immobilisation of a peptide-thiol substrate (SH–$(CH_2)_{10}$–CO–Arg-Val-Arg-Arg) on gold by a two-step sequential procedure: first immerse in the thiol solution, then backfill with 1-octanethiol. This device uses a different approach: all three components (probe-thiol, C6FC, MCH) are co-deposited simultaneously from a single mixed solution (*one-pot deposition*). The Wcisło two-step protocol is not applicable; only the solvent choice (degassed ethanol) is adopted as a starting reference [11].

The one-pot steps are:

1. Immerse polished gold electrode in a mixed ethanolic solution containing probe-thiol, C6FC, and MCH at a total thiol concentration of 3.5 mM for 12–16 h at room temperature. The molar ratios of the three components must be optimised empirically to achieve the target probe coverage.
2. Rinse with ethanol and deionised water; dry under N_2.
3. Optional protein block (BSA, 1% in PBS, 60 min at 4°C) if wound-matrix fouling is anticipated.

The total deposition concentration (3.5 mM) and immersion time (12–16 h) are device-planned values and have not been sourced from any paper in `papers/layer_07/`; both must be optimised by measuring Γ_p on the fabricated electrode.

Langmuir adsorption framework

The equilibrium coverage of a thiol on gold from a solution of molar concentration c is described by the Langmuir isotherm:

$$\Gamma_p = \Gamma_{\max} \cdot \frac{K_L\, c}{1 + K_L\, c} \tag{II.1}$$

where $K_L = k_{\text{ads}}/k_{\text{des}}$ (M^{-1}) is the Langmuir adsorption constant and $\Gamma_{\max}$ is the monolayer saturation coverage.

Saturation coverage. For an analogous Fc-tetrapeptide-PEG_7 system on gold (short peptide on PEG spacer, Fc label, MCH backfill), Adjémian 2010 [12] report saturation at $\Gamma_{\max} \approx 6 \times 10^{-11}$ mol cm^{-2} = 60 pmol cm^{-2} (Experimental Section, p. 10349 of [12]). The Adjémian probe attaches via a disulfide bond (two Au-S contacts) rather than a single Cys-SH; both present a PEG-spacered peptide headgroup that sterically limits packing to well below the alkanethiol limit (930 pmol cm^{-2}, Eq. (I.1)). The $\Gamma_{\max}$ estimate transfers because it is controlled by the steric footprint of the PEG-peptide headgroup, not by the anchor chemistry. Direct coulometric confirmation on the fabricated electrode is required.

Effectively irreversible adsorption. Thiol adsorption on gold is effectively irreversible under typical deposition conditions: the desorption rate constant k_{des} is negligible on the timescale of SAM preparation. Coverage is consequently governed by *kinetic* Langmuir rather than equilibrium concentration. For a single thiol species in isolation, the kinetic Langmuir equation is:

$$\frac{d\Gamma_i}{dt} = k_{\text{ads},i}\, c_i\, (\Gamma_{\max} - \Gamma_i) \tag{II.2}$$

with solution $\Gamma_i(t) = \Gamma_{\max}(1 - e^{-k_{\text{ads},i} c_i t})$, showing that coverage rises toward $\Gamma_{\max}$ at a rate set by exposure time. This is the regime exploited by Adjémian 2010 [12], who deposit a single probe-thiol from a dilute solution and achieve controlled sub-saturating coverages ($\Gamma_p \approx 0.4$–2×10^{-11} mol cm^{-2}) by limiting the immersion time to 30–120 s at 2–5 µM.

For one-pot co-deposition of three competing thiols (probe-thiol, C6FC, MCH), the three equations are coupled: each species competes for the same available sites. The kinetically correct form replaces $(\Gamma_{\max} - \Gamma_i)$ with $(\Gamma_{\max} - \sum_j \Gamma_j)$ for each species i simultaneously. At long times, the coupled kinetic Langmuir converges to the competitive equilibrium isotherm (Eq. (II.3)). The single-component equation (II.2) is therefore not appropriate on its own for describing probe coverage in the one-pot case; the competitive Langmuir framework (Eq. (II.3)) is the primary deposition model for this device.

Working coverage selection. The device requires Γ_p = 2–5 pmol cm^{-2} (steric enzyme-access constraint, probe spacing subsection below), substantially below $\Gamma_{\max}$ = 60 pmol cm^{-2}. In the one-pot regime, the probe fraction $\Gamma_{\text{probe}}/(\Gamma_{\text{probe}} + \Gamma_{\text{C6FC}} + \Gamma_{\text{MCH}})$ is set primarily by the molar ratios of the three thiols in the deposition solution (via Eq. (II.3)), while total surface loading approaches saturation with immersion time. Target probe coverage is therefore achieved by selecting the correct input molar ratio, not by time-limiting the exposure as in the single-component Adjémian approach. Both the molar ratios and the immersion time must be calibrated on the fabricated electrode by coulometric integration of the reductive desorption charge [6].

Three-component competitive Langmuir. With probe-thiol, C6FC, and MCH all present in the deposition solution, the three species compete for gold sites. The competitive Langmuir isotherm for component i is:

$$\theta_i = \frac{K_{L,i}\, c_i}{1 + K_{L,\text{probe}}\, c_{\text{probe}} + K_{L,\text{C6FC}}\, c_{\text{C6FC}} + K_{L,\text{MCH}}\, c_{\text{MCH}}} \tag{II.3}$$

The coverage ratio $\Gamma_{\text{probe}}/\Gamma_{\text{C6FC}}$ therefore depends on $K_{L,\text{probe}}\, c_{\text{probe}} \,/\, K_{L,\text{C6FC}}\, c_{\text{C6FC}}$. The individual Langmuir constants $K_{L,\text{probe}}$, $K_{L,\text{C6FC}}$, and $K_{L,\text{MCH}}$ for the specific thiols used here are not confirmed from any paper in `papers/layer_07/`; the deposition solution molar ratios must therefore be optimised empirically. Once deposited, the individual surface densities Γ_{probe} and Γ_{C6FC} should be verified by reductive desorption coulometry (Widrig [6]) or by integrating the separate CV peaks of MB and Fc in the same post-deposition scan.

Confirmed functional coverage limits

Adjémian et al. (2010) [12] characterised the relationship between surface coverage Γ_0 and cleavage efficiency for a ferrocene (Fc)-labelled tetrapeptide end-grafted on gold via a PEG_7 linker. **Critical applicability note:** the enzymes used in Adjémian 2010 are *trypsin* (bovine pancreas, $M_r \approx 23$ kDa) and *α-thrombin* ($M_r \approx 37$ kDa), not MMP-9 ($M_r \approx 82$–92 kDa). MMP-9 is 3–4 times larger by mass and has a substantially greater hydrodynamic radius. The steric exclusion at the electrode surface will therefore be considerably more severe for MMP-9 than for trypsin at the same probe density, meaning the 10 pmol cm^{-2} functional coverage limit from Adjémian 2010 is *not confirmed* for MMP-9 and may be too permissive. Additionally, the Adjémian assay buffer (50 mM Tris–HCl

pH 8.4, 20 mM $CaCl_2$ for trypsin; 50 mM Tris–HCl pH 8.4, 100 mM NaCl for α-thrombin) differs from wound-fluid conditions (pH ≈7.4).

The Adjémian probe attaches via a disulfide bond (two Au-S contacts) and uses a two-step sequential deposition (probe-thiol then MCH backfill), whereas this device uses a single Cys-SH anchor and one-pot co-deposition of all three thiols. Their key findings (Fig. 2 and text of [12]):

- At saturated undiluted layers ($\Gamma_0 \approx 6 \times 10^{-11}$ mol cm^{-2} = 60 pmol cm^{-2}), cleavage yield drops to ∼50% because chains form compact domains inaccessible to the enzyme.
- For $\Gamma_0 \leq 10^{-11}$ mol cm^{-2} = 10 pmol cm^{-2} (measured with MCH sequential backfill, trypsin at pH 8.4), cleavage yield is $85 \pm 5\%$.
- The working coverage used throughout Adjémian 2010 is $(0.5\text{–}1) \times 10^{-11}$ mol cm^{-2} = 5–10 pmol cm^{-2}.

The device target Γ_p = 2–5 pmol cm^{-2} provides a conservative margin well below the Adjémian trypsin limit; however, the true functional coverage limit for MMP-9 at wound-fluid pH 7.4 is unconfirmed and must be determined on the fabricated electrode. Adjémian 2010 also confirms that surface kinetics follow the Michaelis–Menten two-step mechanism (Layer 9), with enzymatic constants measurable directly from the electrode response.

Parameter	**Status**	**Source**
Deposition protocol	3.5 mM total thiol, 12–16 h (one-pot); all parameters device-planned and unconfirmed from literature	Device-planned
Functional Γ_p limit	Confirmed ($\leq$ 10 pmol cm^{-2})	Adjémian 2010 [12]
k_{cat}, K_m for SGKIPRTLTA	Confirmed (86.4 s^{-1}, 1.3 mM)	Kridel 2001 [10]
Absolute Γ_p on fabricated electrode	**Unconfirmed**: must be measured	Coulometric integration
Surface $k_{\text{eff,surf}}$	**Unconfirmed**: must be measured on device	Layer 9

Table 1: Status of probe-loading parameters. The two unconfirmed items require measurements on the fabricated electrode; they cannot be borrowed from solution or analogous probes.

Probe spacing constraint from enzyme size

Once Γ_p is established from the literature, the average probe spacing is:

$$\ell = (\Gamma_p \cdot N_A)^{-1/2} \tag{II.4}$$

For the sensor to function, MMP-9 (proenzyme molecular weight ≈92 kDa and hydrodynamic radius ≈ 3–4 nm, estimated from MW assuming globular geometry; neither value confirmed from any paper in `papers/layer_07/`) must approach and bind each surface-immobilised peptide. This imposes the requirement $\ell \gtrsim 2 \times r_{\text{MMP-9}} \approx$ 6–8 nm, corresponding to $\Gamma_p \lesssim 3$ pmol cm^{-2}. At higher probe densities, neighbouring probe molecules sterically block enzyme access, reducing the effective cleavage rate and underestimating free MMP-9 activity. This constraint must be confirmed with a cleavage-efficiency vs. coverage measurement on the fabricated electrode.

The following assumptions underlie this section:

1. Probe-thiol, C6FC, and MCH are co-deposited simultaneously; their relative surface coverages are governed by the competitive Langmuir isotherm (Eq. (II.3)).
2. A target of $\Gamma_p \lesssim 3$ pmol cm^{-2} is adopted on steric grounds (enzyme-access requirement); this value is not confirmed from any paper in `papers/layer_07/` and is flagged as Limitation 1.

III. SAM Stability over the Assay Window

Three mechanisms can degrade the SAM during the 60-minute assay: oxidative desorption by H_2O_2, thiol exchange with free thiols in the wound matrix, and electrochemical desorption by the SWV scan potential.

Applicability of alkanethiol literature. All three subsections draw on papers that measured simple n-alkanethiol SAMs. To assess what transfers to the present device, two properties must be distinguished.

Au-S bond thermodynamics (desorption potential, bond energy) are governed by the gold–sulphur bond itself, which is identical for a cysteine-thiol (Cys-SH) and an alkanethiol anchor. The thermodynamic desorption potential and the observation that aqueous exchange is negligible are therefore applicable to the Cys-anchored probe, C6FC, and MCH components of this SAM.

Kinetic rate constants (oxidation rate, exchange rate, desorption peak shape) depend on the SAM packing geometry, chain ordering, and solvent penetration. The probe-thiol here attaches via Cys and extends a PEG_6-peptide chain into solution; MCH ($n = 6$) is structurally closer to the alkanethiol literature but packs alongside the bulky probe rather than in a uniform close-packed layer. Published kinetic rate constants are not quantitatively transferable to this mixed monolayer without experimental confirmation on the fabricated electrode.

The three subsections below use alkanethiol data to establish thermodynamic bounds and qualitative stability arguments, and explicitly flag where only kinetic data would be needed for quantitative prediction.

Oxidative desorption

Schoenfisch and Pemberton characterised alkanethiol SAM oxidation under ambient conditions [5]. Their central finding is that ozone (O_3), not O_2 or UV, is the primary oxidant in air, converting Au–SR to Au^0 + RSO_xH. Under ambient O_3 at 28–60 ppb, a C_{12} SAM on gold lost 36.9% of its total sulphur in 1 h (Table 5 of [5]).

In the present device, the oxidant is H_2O_2 at 10–100 μM in aqueous wound fluid; ozone is absent from the aqueous phase. The relative oxidising power of the two species is reflected in their standard reduction potentials:

$$E^\circ(O_3/O_2) = +2.07 \text{ V}, \qquad E^\circ(H_2O_2/H_2O) = +1.76 \text{ V} \tag{III.1}$$

(Standard IUPAC reduction potentials; no primary paper in `papers/layer_07/` required for these fundamental thermodynamic constants.) The 310 mV thermodynamic advantage of O_3 corresponds to a rate advantage of several orders of magnitude for thiolate oxidation. The Schoenfisch data therefore provide a conservative upper bound: even under the harshest oxidative conditions tested (ambient O_3 in air), SAM loss is approximately 37% in 1 h for a C_{12} chain. Aqueous H_2O_2 at wound-fluid concentrations is a substantially weaker oxidant; oxidative SAM loss over the 60-minute assay is not expected to approach this bound.

No quantitative rate constant k_{ox} for H_2O_2 oxidation of thiol SAMs in solution is available in `papers/layer_07/`. The argument above is comparative only; see Limitation 2.

A separate spontaneous signal-loss mechanism is documented in Adjémian 2010 [12]: an analogous Fc-tetrapeptide-PEG/MCH monolayer on gold (disulfide anchor, two Au-S contacts) showed ∼15–20% signal decrease in blank solution (no enzyme) after 20 min at 37°C (Fig. 1C of [12]), attributed to spontaneous detachment of a small fraction of Fc-peptide chains at elevated temperature. If this device operates at 37°C, a comparable rate of spontaneous probe loss must be characterised on the fabricated electrode before the 60-minute assay window is certified. The Adjémian value is an analogue estimate only: the anchor chemistry (disulfide vs. Cys-SH) and the peptide sequence differ, and the Adjémian assay buffer differs from wound fluid.

Thiol exchange

Schlenoff et al. investigated the exchange of adsorbed alkanethiols on gold by free thiol in solution [7]. The key finding is that the exchange rate shows "very little dependence of rate on thiol concentration," meaning the process is not bimolecular but involves a rate-limiting surface step. In hexane, the pseudo-first-order exchange rate constant is [7]:

$$k_1 = 1.7 \times 10^{-5} \text{ s}^{-1} \quad \text{(hexane, 22°C)} \tag{III.2}$$

The fraction exchanged at time t is $f = 1 - \exp(-k_1 t)$. At $t = 3600$ s (60 min):

$$f = 1 - \exp(-1.7 \times 10^{-5} \times 3600) = 1 - e^{-0.061} \approx 5.9\% \tag{III.3}$$

Equation (III.3) is a strict upper bound: Schlenoff et al. further show that exchange in aqueous conditions is negligible, with essentially no exchange detected even after 125 h in water (Figures 3–6 of [7]). They conclude that desorption is "least pronounced in aqueous systems." Wound fluid is an aqueous matrix; thiol exchange does not contribute meaningfully to SAM loss over the assay window.

Electrochemical desorption

Reductive desorption of alkanethiol SAMs on gold occurs via:

$$\text{Au–SR} + e^- \longrightarrow \text{Au}^0 + \text{RS}^- \tag{III.4}$$

and is only possible when the electrode potential reaches or exceeds the desorption potential E_{des}. Widrig et al. measured E_{des} for a series of ω-alkanethiol SAMs on gold by cyclic voltammetry in 0.5 M KOH, finding a linear dependence on carbon chain length n [6]:

$$E_{\text{des}} = (-0.83 - 0.020\,n)\ \text{V vs Ag/AgCl} \quad (0.5\ \text{M KOH}) \tag{III.5}$$

For MCH ($n = 6$):

$$E_{\text{des}} = -0.83 - 0.020 \times 6 = -0.83 - 0.12 = -0.95\ \text{V} \tag{III.6}$$

Love et al. report that desorption of alkanethiol SAMs from gold typically occurs in the range -1.0 ± 0.25 V vs Ag/AgCl [1], giving a conservative lower bound of -0.75 V. The SWV scan window runs from -0.65 V to $+0.1$ V (Layer 6). The margin between the scan minimum and the conservative lower bound is:

$$\Delta E = E_{\text{des,min}} - E_{\text{scan,min}} = -0.75\ \text{V} - (-0.65\ \text{V}) = -0.10\ \text{V} \tag{III.7}$$

The scan potential is 100 mV more positive than the most conservative desorption estimate; the Widrig value for MCH is 300 mV more negative than the scan limit. Electrochemical desorption does not occur during the assay.

The thermodynamic argument (Au-S bond governs desorption potential) applies to both the Cys-anchored probe-thiol (Ac-Cys-PEG_6–SGKIPRTLTA–PEG_2–Lys(MB)) and MCH: both anchor via the Au-S bond with similar bond energies, placing E_{des} in the same thermodynamic range. However, the probe-thiol and MCH both carry substantial hydrophilic character (PEG_6 polyether, charged peptide headgroup, hydroxyl terminus), and a solvated or disordered SAM has different desorption kinetics than a compact hydrophobic layer: the departing thiolate is stabilised by water, which may lower the effective energy barrier and shift E_{des} toward less negative values relative to the Widrig formula. The actual E_{des} for this mixed monolayer must be confirmed by cyclic voltammetry on the fabricated electrode; see Limitation 4.

The following assumptions underlie this section:

1. The oxidative stability argument uses the Schoenfisch O_3/air benchmark as an upper bound; no quantitative H_2O_2 rate constant is claimed (Limitation 2).
2. Equation (III.2) and the 5.9% bound apply in organic solvent; aqueous exchange is negligible per Schlenoff [7] (Limitation 3).
3. The Love range [1] is used as the primary reference for device-condition E_{des}; the Widrig formula (equation III.5) is supporting data measured specifically in alkaline electrolyte (Limitation 4).

IV. Ca^{2+} Artefact and Ratiometric Cancellation

Mechanism

Wound fluid contains ionised Ca^{2+} at a concentration comparable to serum (∼1–2 mM; no paper in `papers/layer_07/` reports a quantitative wound-fluid Ca^{2+} value: this is an unconfirmed estimate used for the sensitivity calculation below). The probe-thiol (Ac-Cys-PEG_6–SGKIPRTLTA–PEG_2–Lys(MB)) places a PEG_6 block between the Cys anchor and the peptide substrate. Ca^{2+} can coordinate the polyether oxygen atoms of PEG_6 in a crown-ether-like interaction, contracting the spacer and decreasing the average probe-to-gold bridge length by Δd. Electron transfer (ET) from MB to gold through the bridge follows a tunnelling mechanism, so a decrease in bridge length raises the apparent ET rate constant k_{ET} and increases the SWV peak current of *uncleaved* probes.

For the peptide cleavage sensor (signal-off: MB released on cleavage), this Ca^{2+}-induced increase in k_{ET} acts on the uncleaved probe population equally before (I_0) and after ($I(t)$) the assay. The measured quantity is:

$$\frac{\Delta I}{I_0} = \frac{I_0 - I(t)}{I_0} \tag{IV.1}$$

If Ca^{2+} multiplies k_{ET} by a factor R for all uncleaved probes identically, then $I_0 \to R\,I_0$ and $I(t) \to R\,(1-f)\,I_0$ where f is the fraction cleaved. Substituting:

$$\frac{\Delta I}{I_0} = \frac{R\,I_0 - R(1-f)\,I_0}{R\,I_0} = f \tag{IV.2}$$

The Ca^{2+} factor R cancels exactly in the signal-off ratio, provided both I_0 and $I(t)$ are measured in the same buffer (same Ca^{2+} concentration). The signal is inherently self-normalising against Ca^{2+}-induced changes in electron transfer rate, as long as the measurement protocol does not change the buffer between baseline and post-incubation reads.

Tunnelling decay constant

The ET rate constant through a molecular bridge of length d is:

$$k_{ET} = k_{ET}^{0}\,\exp(-\beta\,d) \tag{IV.3}$$

where β is the tunnelling decay constant. Smalley et al. measured β for ferrocene-terminated (Fc) alkanethiol chains tethered to gold in 1.0 M $HClO_4$, by fitting the chain-length dependence of the standard rate constant k° [8]:

$$\beta = 0.94 \pm 0.06\ \text{Å}^{-1} = (9.4 \pm 0.6) \times 10^{9}\ \text{m}^{-1} \tag{IV.4}$$

This measurement was performed on Fc-terminated C_6-to-C_{16} alkanethiol chains; β is a property of the bridge, not of the terminal redox group [8]. For C6FC (Fc–$(CH_2)_6$–SH), the bridge is a C_6 alkyl chain: the Smalley value applies directly. For the probe-thiol (Ac-Cys-PEG_6–SGKIPRTLTA–PEG_2–Lys(MB)), the bridge from Au to MB runs through the Cys residue and the PEG_6 spacer. Tunnelling through polyethylene-glycol chains has a different decay constant from alkyl chains ($\beta_{PEG} < \beta_{alkyl}$ in general), and no paper in `papers/layer_07/` reports β for PEG-based molecular bridges. The value $\beta = 0.94$ Å^{-1} is used here as a provisional upper bound for the Ca^{2+} sensitivity calculation below; the actual fold change in k_{ET} is unconfirmed and is flagged as Limitation 5.

Sensitivity of k_{ET} to bridge contraction

If Ca^{2+} coordination contracts the PEG spacer by Δd, the fold increase in k_{ET} follows from equation (IV.3):

$$\frac{k_{ET}^{Ca}}{k_{ET}^{0}} = \exp(\beta\,\Delta d) \tag{IV.5}$$

For a representative contraction of $\Delta d = 0.2$ nm (one carbon–carbon bond displacement, used as a sensitivity bound since Δd is not measured for this architecture; see Limitation 5):

$$\begin{aligned}\frac{k_{ET}^{Ca}}{k_{ET}^{0}} &= \exp(9.4 \times 10^{9}\ \text{m}^{-1} \times 0.2 \times 10^{-9}\ \text{m}) \\ &= \exp(1.88) \approx 6.5\end{aligned} \tag{IV.6}$$

Both the active and control electrodes carry the same probe-thiol/C6FC/MCH SAM and are exposed simultaneously to the same wound fluid, so this fold increase applies identically to both.

Residual error from probe density change after cleavage

Equation (IV.2) assumes that Ca^{2+} affects uncleaved probes with the same k_{ET} before and after the assay. This holds exactly if the cleaved probe stub (the fragment remaining on the electrode after MB release) has negligible electron transfer compared with the intact probe. If the stub retains some residual MB (partial cleavage) or if cleavage changes the spacer geometry in a way that modifies the Ca^{2+} response of neighbouring probes, a second-order residual arises. The magnitude of this residual depends on the specific peptide architecture and is not quantifiable without a calibration measurement. It is expected to be small (second-order in f and in $\delta k_{ET,stub}$), but experimental confirmation that $\Delta I/I_0$ is Ca^{2+}-independent in the assay buffer is recommended.

Limitations of this model

1. **Deposition parameters are a starting point; molar ratios and absolute Γ_p on the fabricated electrode must be measured.** This device uses one-pot co-deposition of all three thiols (3.5 mM total thiol, 12–16 h in ethanol) rather than any literature two-step procedure. The 3.5 mM concentration, 12–16 h immersion time, and molar ratios of probe-thiol, C6FC, and MCH are not sourced from any paper in `papers/layer_07/` and must all be optimised empirically on the fabricated electrode. Adjémian 2010 [12] reports $85 \pm 5\%$ cleavage yield for $\Gamma_0 \leq 10$ pmol cm^{-2}, but this was established with trypsin ($M_r \approx 23$ kDa) at pH 8.4 with 20 mM $CaCl_2$ — conditions very different from MMP-9 ($M_r \approx 82$–92 kDa) in wound fluid at pH 7.4. The true functional coverage limit for MMP-9 in wound-fluid conditions is unconfirmed from any paper in `papers/layer_07/` and must be determined experimentally. What remains unconfirmed is the *actual* Γ_p achieved on the fabricated SGKIPRTLTA–MB electrode. This must be measured directly by reductive desorption coulometry (method: Widrig [6]) or equivalent before Layer 8 and Layer 9 outputs can be quantified.
2. **No quantitative k_{ox} for H_2O_2 oxidation of thiol SAMs.** Schoenfisch and Pemberton [5] measure SAM oxidation by O_3 in air, not by H_2O_2 in solution. No paper in `papers/layer_07/` provides k_{ox} for aqueous H_2O_2. The oxidative stability argument is comparative only; experimental confirmation of SAM integrity after 60 min in wound-fluid H_2O_2 is required before the assay window can be certified.
3. **Thiol exchange rate confirmed in hexane only.** Schlenoff et al. [7] measure $k_1 = 1.7 \times 10^{-5}$ s^{-1} in hexane and demonstrate negligible aqueous exchange qualitatively; a quantitative rate constant for exchange by Cys or GSH in phosphate-buffered wound fluid at pH 7.4 is not available.
4. **E_{des} measured in 0.5 M KOH on hydrophobic alkanethiols; both conditions differ from this device.** The Widrig formula (equation III.5) was determined at pH ≈ 13.7 [6]. At wound-fluid pH 7.4 the Nernst equation predicts that E_{des} shifts approximately $+(13.7 - 7.4) \times 0.059 \approx +0.37$ V toward positive values (because RS^- is protonated at pH 7.4), placing E_{des} near $-0.95 + 0.37 = -0.58$ V. This is only 70 mV positive of the scan minimum of -0.65 V, compared with the 300 mV margin in alkaline electrolyte. The Love range [1] (-1.0 ± 0.25 V, conditions unspecified) is used as the primary working reference. A second applicability concern arises from molecular structure: Widrig's formula was measured on hydrophobic n-alkanethiols with no polar headgroups and no solvent penetration into the SAM interior. The probe-thiol and MCH in this device have substantial hydrophilic character (PEG_6 polyether, peptide headgroup, hydroxyl terminus of MCH). Greater water exposure and structural disorder lower the thermodynamic barrier to desorption and may shift E_{des} toward less negative values relative to the Widrig formula. The formula is not transferable to this mixed peptide-thiol/MCH monolayer; experimental determination of E_{des} in the assay buffer on the fabricated electrode is required before the scan window is certified safe.
5. **Ca^{2+} artefact model has three unconfirmed inputs.**
   (a) *Protocol.* Equation (IV.2) shows that the Ca^{2+} artefact cancels in $\Delta I/I_0$ only if I_0 and $I(t)$ are measured in the same buffer. If I_0 is taken in calibration buffer (no Ca^{2+}) and $I(t)$ in wound fluid (with Ca^{2+}), the cancellation fails. Protocol design must ensure both measurements are in the same ionic environment.
   (b) *PEG spacer Ca^{2+} affinity.* The Ca^{2+}-coordination mechanism assumes the PEG_6 block contracts upon Ca^{2+} binding. Whether a PEG_6 spacer of this length binds Ca^{2+} with meaningful affinity in the wound-fluid ionic background is not confirmed from any paper in `papers/layer_07/`; Δd is unconfirmed.
   (c) *Tunnelling decay constant for PEG bridge.* Smalley 2003 [8] measures $\beta = 0.94$ Å^{-1} for C_6–C_{16} alkyl chains. The probe-thiol bridge from Au to MB runs through Cys and PEG_6; the β for PEG chains is smaller than for alkyl chains and is not confirmed in `papers/layer_07/`. The fold change in k_{ET} estimated in equation (IV.6) is therefore an upper bound.

References

All papers in `electrochemical_device/papers/layer_07/`.

[1] Love JC, Estroff LA, Kriebel JK, Nuzzo RG, Whitesides GM (2005). Self-assembled monolayers of thiolates on metals as a form of nanotechnology. *Chem Rev* 105(4):1103–1169. **love2005.pdf** – Γ_{max} cross-check (≈ 750 pmol cm^{-2}, p. 1108), $E_{des} = -1.0 \pm 0.25$ V (p. 1122) confirmed from text.

[2] Herne TM, Tarlov MJ (1997). Characterization of DNA probes immobilized on gold surfaces. *J Am Chem Soc* 119(38):8916–8920. **herne1997.pdf** – *DNA-thiol specific. KH_2PO_4 buffer requirement, log-time deposition kinetics, and MCH displacement of non-thiolated DNA all pertain to HS-ssDNA with 25 charged phosphate groups per strand. These results do not transfer to the peptide-thiol probe and are not used for deposition protocol in this layer.*

[3] Steel AB, Herne TM, Tarlov MJ (1998). Electrochemical quantitation of DNA immobilized on gold. *Anal Chem* 70(22):4670–4677. **steel1998.pdf** – *DNA-thiol specific. The functional coverage limit (4×10^{12} molecules cm^{-2}) was determined by hybridisation steric hindrance. Enzyme-cleavage steric limits for MMP-9 are different and*

not covered by this paper.

[4] Petrovykh DY, Kimura-Suda H, Whitman LJ, Tarlov MJ (2003). Quantitative analysis and characterization of DNA immobilized on gold. *J Am Chem Soc* 125(17):5219–5226. **petrovykh2003.pdf** – *DNA-thiol specific. Log-linear adsorption kinetics and the non-Langmuir conclusion apply to HS-ssDNA. Peptide-thiol adsorption kinetics are not covered by this paper.*

[5] Schoenfisch MH, Pemberton JE (1998). Air stability of alkanethiol self-assembled monolayers on silver and gold surfaces. *J Am Chem Soc* 120(18):4502–4513. **schoenfisch1998.pdf** – O_3 as primary oxidant confirmed from text; C_{12} SAM 36.9% sulphur loss in 1 h confirmed from Table 5. Paper does not give k_{ox} for aqueous H_2O_2; value attributed to this paper in `sam_stability.py` is not confirmed and has been removed.

[6] Widrig CA, Chung C, Porter MD (1991). The electrochemical desorption of *n*-alkanethiol monolayers from polycrystalline Au and Ag electrodes. *J Electroanal Chem* 310(1–2):335–359. **widrig1991.pdf** – $\Gamma_{max} = 9.3 \times 10^{-10}$ mol cm^{-2} (p. 351) confirmed; E_{des} formula (equation III.5) confirmed from text. Measured in 0.5 M KOH; pH correction noted in Limitation 4.

[7] Schlenoff JB, Li M, Ly H (1995). Stability and self-exchange in alkanethiol monolayers. *J Am Chem Soc* 117(50):12528–12536. **schlenoff1995.pdf** – $k_1 = 1.7 \times 10^{-5}$ s^{-1} in hexane confirmed from Table 3; concentration-independence of exchange rate confirmed from text; negligible aqueous exchange confirmed from Figures 3–6. Bimolecular rate model in `sam_stability.py` is not supported by this paper and has been removed.

[8] Smalley JF, Feldberg SW, Chidsey CED, Linford MR, Newton MD, Liu Y-P (2003). The kinetics of electron transfer through ferrocene-terminated alkanethiol monolayers on gold. *J Am Chem Soc* 125(7):2004–2013. **smalley2003.pdf** – $\beta = 0.94 \pm 0.06$ Å^{-1} confirmed from text (p. 2009). Measured using ferrocene (Fc) redox probe; value transfers to methylene-blue system because β is a property of the alkyl bridge, confirmed from discussion in [8].

[9] Becka AM, Miller CJ (1993). Electrochemistry at ω-hydroxy thiol coated electrodes. 3. Voltage independence of the electron tunneling barrier and measurements of redox kinetics at large overpotentials. *J Phys Chem* 97(23):6233–6239. **becka1993.pdf** – provides context for tunnelling through alkanethiol SAMs; the $\beta = 0.94$ Å^{-1} value used in this layer is from Smalley 2003 [8], not Becka 1993.

[10] Kridel SJ, Chen E, Kotra LP, Howard EW, Mobashery S, Smith JW (2001). Substrate hydrolysis by matrix metalloproteinase-9. *J Biol Chem* 276(23):20572–20578. DOI: 10.1074/jbc.M100900200. **PIIS0021925819405115.pdf** – $k_{cat} = 86.4$ s^{-1}, $K_m = 1.3$ mM, $k_{cat}/K_m = 67{,}500$ M^{-1}s^{-1} for peptide A11 (SGKIPRTLTA) confirmed from Table II. These solution-phase values are an upper bound for the surface $k_{eff,surf}$.

[11] Wcisło A, Małuch I, Niedziałkowski P, Ossowski T, Prahl A (2021). Label-free electrochemical test of protease interaction with a peptide substrate modified gold electrode. *Chemosensors* 9(8):199. DOI: 10.3390/chemosensors9080199. **10.3390@chemosensors9080199.pdf** – deposition protocol for peptide-thiol on gold (5 mM EtOH, 20 h, backfill with alkanethiol, BSA block) confirmed from Section 2.3.

[12] Adjémian J, Anne A, Cauet G, Demaille C (2010). Cleavage-sensing redox peptide monolayers for the rapid measurement of the proteolytic activity of trypsin and α-thrombin enzymes. *Langmuir* 26(12):10347–10356. DOI: 10.1021/la100397g. **10.1021@la100397g.pdf** – cleavage yield >80% for $\Gamma_0 \leq 10$ pmol cm^{-2} with MCH backfill confirmed from Fig. 2 and text (p. 10351); working coverage (0.5–1) $\times 10^{-11}$ mol cm^{-2} confirmed; saturation $\Gamma_{max} \approx 60$ pmol cm^{-2} confirmed from Experimental Section (p. 10349); Fc formal potential 150–155 mV vs. SCE confirmed from p. 10350. Architecture (Fc-peptide-PEG tethered on Au, MCH backfill) is analogous to this device; anchor is disulfide (two Au-S contacts) versus Cys-SH (single Au-S) in this device.

[13] Sfragano PS, Moro G, Polo F, Palchetti I (2021). The role of peptides in the design of electrochemical biosensors for clinical diagnostics. *Biosensors* 11(8):246. DOI: 10.3390/bios11080246. **sfragano2021.pdf** – review of peptide-based electrochemical biosensors; confirms PEG linker strategies, cysteine-gold anchoring, and protease cleavage signal-off mechanisms consistent with this device architecture.